

EXPERIMENT – 6

Title: To Use Oracle Functions viz. Aggregate, Numeric, Conversion, and String Functions

Aim / Objective

To study and implement various Oracle built-in functions including **Aggregate Functions, Numeric Functions, Conversion Functions, and String Functions** for data retrieval and manipulation.

Theory

Oracle provides a large number of built-in functions that simplify data processing and manipulation. These functions help users perform calculations, convert data types, manipulate strings, and summarize data.

Oracle functions are broadly classified into:

- Aggregate Functions
- Numeric Functions
- Conversion Functions
- String Functions

These functions improve query efficiency and reduce the complexity of SQL statements.

Types of Oracle Functions

1. Aggregate Functions

Aggregate functions perform calculations on a group of rows and return a single value.

Common aggregate functions:

- COUNT()
- SUM()
- AVG()
- MAX()
- MIN()

2. Numeric Functions

Numeric functions perform mathematical operations on numeric values.

Common numeric functions:

- ROUND()
- TRUNC()
- MOD()

- ABS()
- CEIL()
- FLOOR()
- POWER()

3. Conversion Functions: Conversion functions convert one datatype into another.

Common conversion functions:

- TO_CHAR()
- TO_DATE()
- TO_NUMBER()

4. String Functions

String functions manipulate character data.

Common string functions:

- UPPER()
- LOWER()
- INITCAP()
- LENGTH()
- SUBSTR()
- CONCAT()
- INSTR()
- REPLACE()
- TRIM()

Learning Objectives

After completing this experiment, students will be able to:

- Use Oracle aggregate functions.
- Perform mathematical calculations using numeric functions.
- Convert data types using conversion functions.
- Manipulate strings using string functions.
- Apply Oracle functions in SQL queries.

Software Requirements

- Oracle Database
- Oracle SQL Developer / SQL*Plus

Algorithm / Procedure

1. Open Oracle SQL Developer.
2. Connect to the Oracle database.
3. Create the Employee table.
4. Insert sample records.
5. Execute SQL queries using Oracle functions.
6. Observe the output.
7. Stop.

Program

Step 1 : Create Employee Table

```
CREATE TABLE Employee
(
EmpID NUMBER PRIMARY KEY,
EmpName VARCHAR2(30),
Department VARCHAR2(30),
Salary NUMBER(10,2),
JoiningDate DATE
);
```

Step 2 : Insert Sample Records

```
INSERT INTO Employee VALUES
(101,'Rahul','IT',35000,TO_DATE('15-01-2022','DD-MM-YYYY'));
```

```
INSERT INTO Employee VALUES
(102,'Priya','HR',42000,TO_DATE('10-02-2021','DD-MM-YYYY'));
```

```
INSERT INTO Employee VALUES
(103,'Aman','Finance',39000,TO_DATE('20-03-2023','DD-MM-YYYY'));
```

```
INSERT INTO Employee VALUES
(104,'Sneha','Marketing',45000,TO_DATE('05-07-2020','DD-MM-YYYY'));
```

```
COMMIT;
```

Aggregate Functions

Program 1 : COUNT()

```
SELECT COUNT(*) AS Total_Employees
```

FROM Employee;

Expected Output

Total Employees

4

Program 2 : SUM()

```
SELECT SUM(Salary)
AS Total_Salary
FROM Employee;
```

Expected Output

Total Salary

161000

Program 3 : AVG()

```
SELECT AVG(Salary)
AS Average_Salary
FROM Employee;
```

Expected Output

Average Salary

40250

Program 4 : MAX()

```
SELECT MAX(Salary)
FROM Employee;
```

Expected Output

MAX(Salary)

45000

Program 5 : MIN()

```
SELECT MIN(Salary)
FROM Employee;
```

Expected Output

MIN(Salary)

35000

Numeric Functions

Program 6 : ROUND()

```
SELECT ROUND(1234.567,2)
FROM Dual;
```

Expected Output

ROUND

1234.57

Program 7 : TRUNC()

```
SELECT TRUNC(1234.567,2)
FROM Dual;
```

Expected Output

TRUNC

1234.56

Program 8 : MOD()

```
SELECT MOD(25,4)
FROM Dual;
```

Expected Output

MOD

1

Program 9 : ABS()

```
SELECT ABS(-125)
FROM Dual;
```

Expected Output

ABS

125

Program 10 : POWER()

```
SELECT POWER(5,2)
```

FROM Dual;

Expected Output

POWER

25

Conversion Functions

Program 11 : TO_CHAR()

```
SELECT TO_CHAR(JoiningDate,'DD-MON-YYYY')  
FROM Employee;
```

Program 12 : TO_NUMBER()

```
SELECT TO_NUMBER('1500')  
FROM Dual;
```

Program 13 : TO_DATE()

```
SELECT TO_DATE('15-08-2024','DD-MM-YYYY')  
FROM Dual;
```

String Functions

Program 14 : UPPER()

```
SELECT UPPER(EmpName)  
FROM Employee;
```

Program 15 : LOWER()

```
SELECT LOWER(EmpName)  
FROM Employee;
```

Program 16 : INITCAP()

```
SELECT INITCAP('oracle database')  
FROM Dual;
```

Expected Output

INITCAP

Oracle Database

Program 17 : LENGTH()

```
SELECT LENGTH(EmpName)  
FROM Employee;
```

Program 18 : SUBSTR()

```
SELECT SUBSTR(EmpName,1,3)
FROM Employee;
```

Expected Output

SUBSTR

Rah

Pri

Ama

Sne

Program 19 : CONCAT()

```
SELECT CONCAT(EmpName,' Employee')
FROM Employee;
```

Program 20 : REPLACE()

```
SELECT REPLACE('Oracle Database','Database','SQL')
FROM Dual;
```

Expected Output

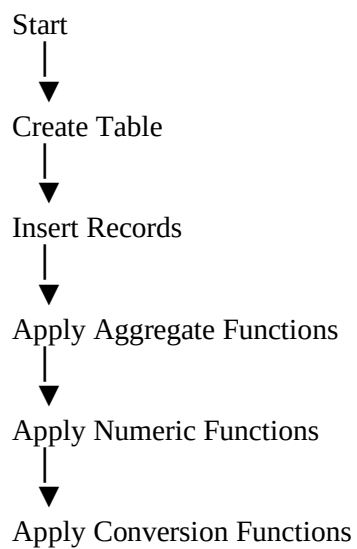
REPLACE

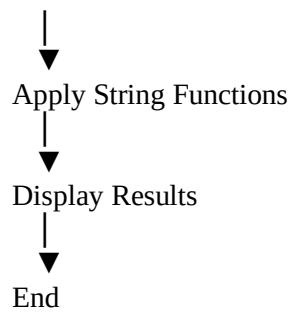
Oracle SQL

Program 21 : TRIM()

```
SELECT TRIM(' Oracle ')
FROM Dual;
```

Flowchart





Explanation

Aggregate Functions: Perform calculations on multiple rows and return a single result.

Numeric Functions: Perform mathematical operations on numbers.

Conversion Functions: Convert one data type into another.

String Functions: Manipulate and process character strings.

Observation

- Aggregate functions calculated totals, averages, maximum, and minimum values successfully.
- Numeric functions performed mathematical operations correctly.
- Conversion functions converted numbers, dates, and strings accurately.
- String functions manipulated text data effectively.

Result

Oracle Aggregate, Numeric, Conversion, and String functions were successfully implemented. The functions performed calculations, datatype conversions, and string manipulations efficiently, demonstrating their usefulness in database applications.

Applications

- Payroll Systems
- Banking Applications
- Hospital Management Systems
- Student Information Systems
- Sales Reporting
- Inventory Management
- Human Resource Management Systems

Advantages

- Simplifies complex SQL queries.

- Reduces coding effort.
- Improves query performance.
- Supports efficient data analysis.
- Enhances report generation.

Viva Questions

1. What are Oracle built-in functions?
2. What is the difference between single-row and aggregate functions?
3. What is the purpose of the COUNT() function?
4. What is the difference between ROUND() and TRUNC()?
5. What is the use of the MOD() function?
6. What is the purpose of the TO_CHAR() function?
7. What is the difference between TO_DATE() and TO_CHAR()?
8. What is the purpose of the UPPER() function?
9. What is the difference between SUBSTR() and INSTR()?
10. What is the purpose of the DUAL table in Oracle?

Lab Assignment

1. Find the total salary of all employees.
2. Display the highest and lowest employee salary.
3. Calculate the average salary of employees.
4. Convert the current system date into the format **DD-MON-YYYY**.
5. Display employee names in uppercase and lowercase.
6. Display the first four characters of employee names.
7. Replace the word **Oracle** with **SQL** in a given string.
8. Find the remainder when 75 is divided by 8 using the MOD() function.
9. Compare Aggregate Functions and Single-Row Functions with suitable examples.
10. Write SQL queries using at least five Oracle built-in functions.

Precautions

- Ensure the table contains valid data before applying functions.

- Use aggregate functions with appropriate columns.
- Specify the correct format model while using TO_CHAR() and TO_DATE().
- Use string functions only with character data.
- Verify the output after executing each SQL statement.

Learning Outcome

After completing this experiment, students will be able to:

- Use Oracle aggregate functions to summarize data.
- Apply numeric functions for mathematical calculations.
- Convert data types using conversion functions.
- Manipulate character data using string functions.
- Write efficient SQL queries using Oracle built-in functions for real-world database applications.